J2 Vector magnitude and direction

Name: ______________________

Class: ______________________

Date: ______________________

Time: **38 minutes**

Marks: **31 marks**

Comments:

Q1.

Two forces $\mathbf{F}_1 = (3\mathbf{i} + 4\mathbf{j})$N and $\mathbf{F}_2 = (6\mathbf{i} - 8\mathbf{j})$N, act on a particle. The resultant of these two forces is **F**. The unit vectors **i** and **j** are perpendicular.

(a) Find **F**.

(2)

(b) Find the magnitude of **F.**

(2)

(c) Find the acute angle between **F** and the unit vector **i.**

(3)

(Total 7 marks)

Q2.

A particle moves on a horizontal plane subject to the following three forces:

$\mathbf{F}_1 = (2\mathbf{i} + 3\mathbf{j})$N, $\mathbf{F}_2 = (6\mathbf{i} - 8\mathbf{j})$N and $\mathbf{F}_3 = (-2\mathbf{i} + 10\mathbf{j})$N

The unit vectors **i** and **j** are perpendicular and lie in the horizontal plane.

Do not include any other forces in your calculations.

(a) Find the resultant force on the particle.

(2)

(b) The mass of the particle is 8 kg. Find the acceleration of the particle.

(1)

(c) At time $t = 0$, the particle is at the origin and has velocity $(3\mathbf{i} - 2\mathbf{j})\text{ms}^{-1}$.

(i) Find the velocity of the particle when $t = 40$ seconds.

(2)

(ii) Find the distance of the particle from the origin when $t = 40$ seconds.

(5)

(Total 10 marks)

Q3.

At time $t = 0$, a particle is at the origin and moving with velocity $(4\mathbf{i} + 2\mathbf{j})$ m s^{-1}.
When $t = 10$ seconds the position vector of the particle is $(44\mathbf{i} + 28\mathbf{j})$ metres.
The particle is moving with constant acceleration.

(a) Find the acceleration of the particle.

(4)

(b) Find the position vector of the particle at time t.

(3)

(c) At time $t = T$, the position vector of the particle is $(96\mathbf{i} + 72\mathbf{j})$ metres.

(i) Find T.

(4)

(ii) Find the speed of the particle at this time.

(3)

(Total 14 marks)